

(ii) On Fig. 3.2, sketch the variation with F of the spring constant k of the wire.



Fig. 3.2

[1]

(d) Copper has a resistivity of $1.8 \times 10^{-8} \Omega \text{ m}$ and a Young modulus of $1.3 \times 10^{11} \text{ Pa}$.

A copper wire of diameter 1.6 mm has a resistance of 0.034Ω .

(i) Show that the length of the wire is 3.8 m.

[1]

(ii) Use the equation in (b)(ii) to determine the spring constant of the wire.

spring constant = N m^{-1} [2]

[Total: 9]

